



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- I-A) What is the difference between long-term scheduler and short-term scheduler?**
- I-B) What are the advantages of using Semaphores?**
- I-C) Following code is a solution of Dining-Philosopher problem:**

```
while (true){  
    wait (chopstick[i] );  
    wait (chopstick[ (i + 1) % 5] );  
  
    /* eat for a while */  
    signal (chopstick[i] );  
    signal (chopstick[ (i + 1) % 5] );  
    while  
        /* think for a while */  
}
```

Is there any problem in this code? Can we improve this?

- I-D) What is context-switch? What does it mean by dispatch-latency?**
- I-E) What is Race Condition? Explain with the help of an example.**
- I-F) Consider a logical address space of 64 pages of 2048 bytes each, mapped onto a physical memory of 32 frames.**
 - a. How many bits are there in the logical address?**
 - b. How many bits are there in the physical address?**
 - c. How many bits needed for offset?**

Q.2. Answer the following questions.

(3×10=30)

- II-A) Consider a system where size of physical frame is 16 bytes. The virtual addresses of the process are 6 bits in size. The program generates the following 20 virtual address references as it runs. Convert these virtual addresses into form of page number and offset.

Virtual address	Page #	Offset
0		
1		
20		
2		
20		
21		
32		
31		
0		
60		
1		
17		
18		
32		
31		

- II-B) Consider a system with four types of resources (Tape drives, Plotters, Scanners and CD-ROMs). There are 4,2,3 and 1 instances of these resources. Currently system has only 2 Tape drives and 1 Plotter available.

Following table shows Allocation of resources to three processes currently running:

	Tape Drives	Plotters	Scanners	CD-ROMs
P1	0	0	1	0
P2	2	0	0	1
P3	0	1	2	0

Maximum allocation required by each process is given below:

	Tape Drives	Plotters	Scanners	CD-ROMs
P1	3	0	2	1
P2	2	1	3	1
P3	1	2	2	0

Use Banker's algorithm to check whether the system is in Safe-State or not?

- II-C) What is the difference between Task-Level parallelism and Data-Level parallelism? Give examples to explain both.